Model selection

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Texas Suffers From A Doctor Shortage

By JONATHAN BAKER • NOV 1, 2017

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When it comes to having a high ratio of doctors to citizens, the State of Texas ranks near the bottom. In fact, as *The Dallas Morning News* reports, 43 states have a higher proportion of primary care physicians to residents than Texas.

And West Texas suffers from a lack of doctors more than other parts of the state. There are 80 counties in Texas with five or fewer practicing doctors - many in West Texas. Thirty-five Texas counties have no doctors at all.

The Texas counties data set

- All 168 Texas counties with 10,000+ residents.
- The response: **physicians per 10,000 people**, which runs from 0.0 to 12.9 across counties.
- Ten candidate predictors describing each county's land, people, and economy.

Our question: which of these factors predict how many doctors a county has?

Ten candidate predictors

`LandArea`	area in square miles	`PctOver65`	percent seniors
`PctRural`	percent rural land	`PctPoverty`	percent in poverty
`MedianIncome`	median household income	`PctUninsured`	percent uninsured
`Population`	population	`PctSomeCollege`	percent with some college
`PctUnder18`	percent children	`PctUnemployed`	percent unemployed

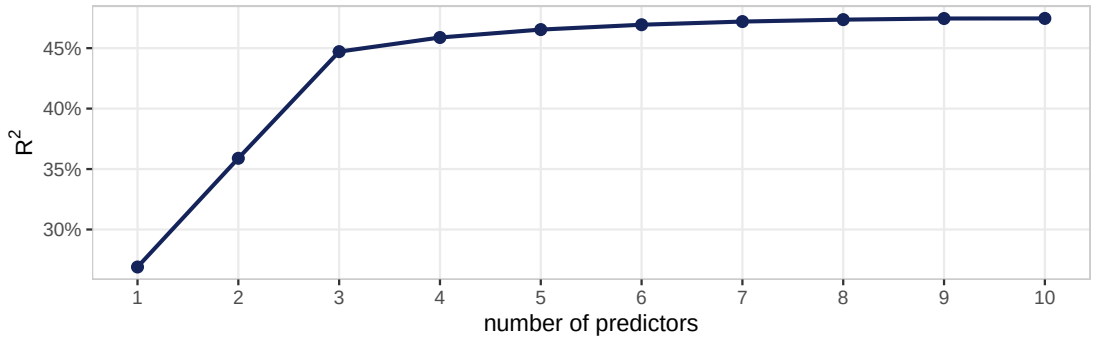
Which would you expect to matter? Which would you leave out?

With ten predictors there are 1,023 possible regressions.

How do we know which model is best?

Why not just use R^2?

The best model of each size, by R^2:



R^2 **never goes down** when you add a predictor. Maximize it and you always keep everything.

Suppose we want to predict the Democratic candidate's share of the vote in a presidential election...

THE PROBLEM WITH STATEMENTS LIKE

"NO <PARTY> CANDIDATE HAS WON THE ELECTION WITHOUT <STATE>"

OR

"NO PRESIDENT HAS BEEN REELECTED UNDER <CIRCUMSTANCES>"

★ UPDATED FOR 2020 ★

1788...
NO ONE HAS BEEN ELECTED PRESIDENT BEFORE.

...BUT WASHINGTON WAS.

1792...
NO INCUMBENT HAS EVER BEEN REELECTED.

...UNTIL WASHINGTON.

1796...
NO ONE WITHOUT FALSE TEETH HAS BECOME PRESIDENT.

...BUT ADAMS DID.

1800...
NO CHALLENGER HAS BEATEN AN INCUMBENT.

...BUT JEFFERSON DID.

1804...
NO INCUMBENT HAS BEATEN A CHALLENGER.

...UNTIL JEFFERSON.

1808...
NO CONGRESSMAN HAS EVER BECOME PRESIDENT.

...UNTIL MADISON.

1812...
NO ONE CAN WIN WITHOUT NEW YORK.

...BUT MADISON DID.

1816...
NO CANDIDATE WHO DOESN'T WEAR A WIG CAN GET ELECTED.

...UNTIL MONROE WAS.

1820...
NO ONE WHO WEARS PANTS INSTEAD OF BREECHES CAN BE REELECTED.

...BUT MONROE WAS.

1824...
NO ONE HAS EVER WON WITHOUT A POPULAR MAJORITY.

...J.Q. ADAMS DID.

1828...
ONLY PEOPLE FROM MASSACHUSETTS AND VIRGINIA CAN WIN.

...UNTIL JACKSON DID.

1832...
THE ONLY PRESIDENTS WHO GET REELECTED ARE VIRGINIANS.

...UNTIL JACKSON.

1836...
NEW YORKERS ALWAYS LOSE.

...UNTIL VAN BUREN.

1840...
NO ONE OVER 65 HAS WON THE PRESIDENCY.

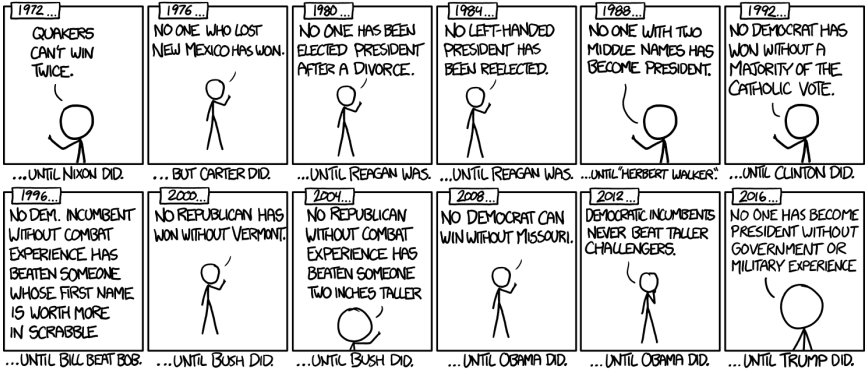
...UNTIL HARRISON DID.

1844...
NO ONE WHO'S LOST HIS HOME STATE HAS WON.

...BUT POLK DID.

1848...
AS GOES MISSISSIPPI, SO GOES THE NATION.

...UNTIL 1848.



CONGRATULATIONS TO PRESIDENT-ELECT JOE BIDEN FOR BREAKING THE WEBSITE CURSE!

Every one of these raises R^2

- Whether the candidate has won Missouri

Every one of these raises R^2

- Whether the candidate has won Missouri
- The height difference between the candidates

Every one of these raises R^2

- Whether the candidate has won Missouri
- The height difference between the candidates
- Whether the candidate has combat or government experience

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- Whether the candidate has won Missouri
- The height difference between the candidates
- Whether the candidate has combat or government experience
- Whether the candidate has two middle names

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- Whether the candidate has won Missouri
- The height difference between the candidates
- Whether the candidate has combat or government experience
- Whether the candidate has two middle names
- Whether the candidate is left-handed

Every one of these raises R^2

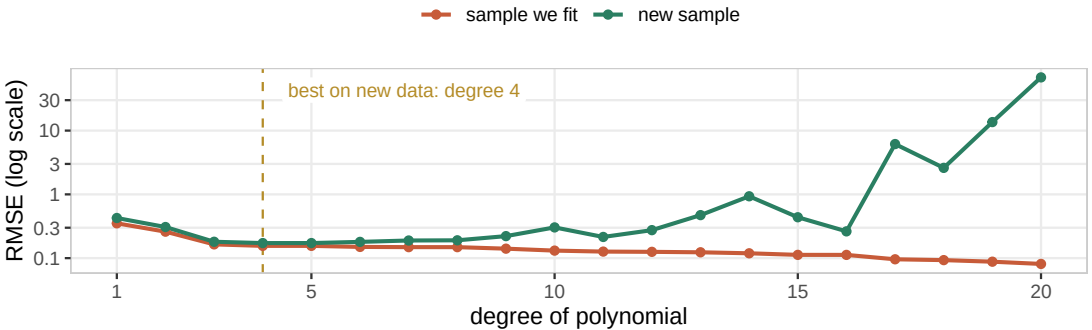
- Whether the candidate has won Missouri
- The height difference between the candidates
- Whether the candidate has combat or government experience
- Whether the candidate has two middle names
- Whether the candidate is left-handed
- Whether the candidate's opponent has won more games of Scrabble

Every one of these raises R^2

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How much would you trust a model with dozens of variables like these?

Recall: overfitting, in one picture



Polynomials fit to a noisy wave: past degree 4, we fit the noise.

More predictors in a regression behave the same way.

Adjusted R^2

Adjusted R^2 penalizes R^2 using the number of predictors k and cases n:

$$\text{Adjusted } R^2 = 1 - (1 - R^2)\,\frac{n-1}{n-k-1}$$

- A new predictor must raise R^2 by enough to beat the larger k, or adjusted R^2 falls.
- It keeps roughly the "share of variance explained" reading of R^2.
- Higher is better.

AIC: the Akaike information criterion

$$\text{AIC} = 2k - 2\log(L) \quad \overset{\text{linear regression}}{\Longrightarrow} \quad \text{AIC} = n\log\left(\frac{\text{RSS}}{n}\right) + 2k$$

$$\underbrace{n\log(\text{RSS}/n)}_{\textbf{fit}} \quad + \quad \underbrace{2k}_{\textbf{complexity}}$$

falls as predictors are added | rises by 2 per parameter

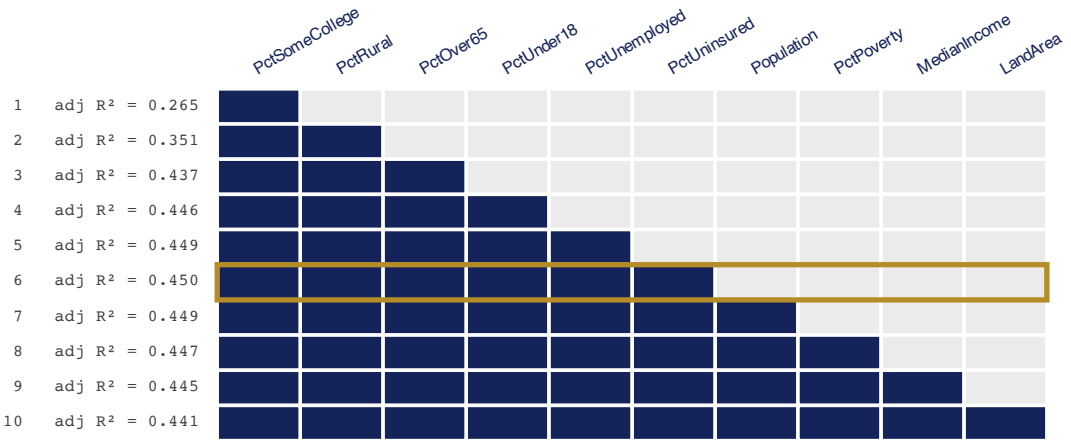
- k counts parameters (predictors plus the intercept). Only **differences** in AIC mean anything.
- A predictor earns its place only if it improves the fit by more than the penalty it costs. Lower is better.

Best subsets regression

- Computers are fast. Let R fit all $2^k - 1$ models with at least one of the k predictors.
- For each model size, keep the best one; then compare the winners by adjusted R^2 or AIC.
- The catch: every new predictor **doubles** the work. 10 predictors means 1,023 models; 40 means about a trillion.

```
library(leaps)
subsets <- regsubsets(PhysiciansPer10000 ~ ., data=counties, nvmax=10)
summary(subsets)$adjr2
```

The winner at every size



Each row is the best model of that size. Adjusted R^2 peaks at 6 predictors (gold), then falls.

Backward stepwise regression

1. Start with the **full** model containing every predictor.
2. Try removing each predictor in turn, and compute the AIC of each smaller model.
3. If any removal lowers the AIC, make the best one and go back to step 2. If none does, stop.

About $k^2/2$ fits instead of 2^k: with 40 predictors, roughly 800 instead of a trillion.

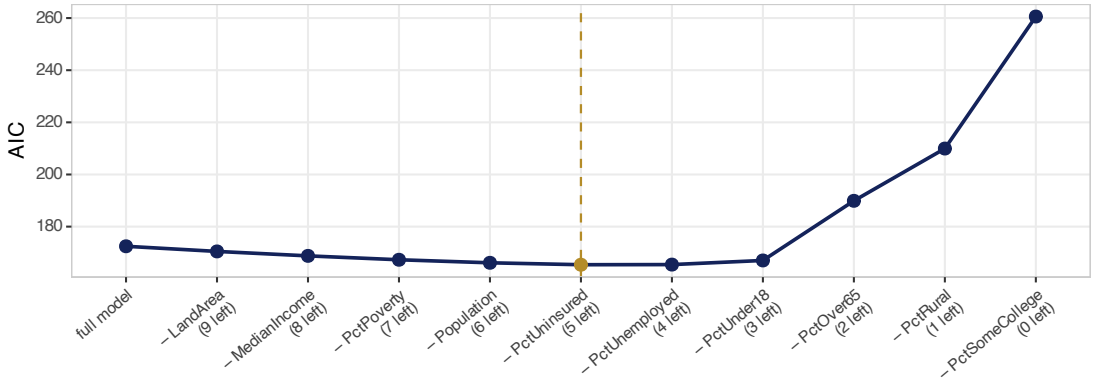
Running it in R

```
full <- lm(PhysiciansPer10000 ~ ., data=counties)
step.model <- step(full, trace=0)
formula(step.model)

PhysiciansPer10000 ~ PctRural + PctUnder18 + PctOver65 + PctSomeCollege +
    PctUnemployed
```

`step()` drops 5 predictors and keeps 5. Set `trace=1` to watch every step.

Watching the elimination



Read left to right: each step removes one predictor. AIC bottoms out at 5 predictors, which is where `step()` stops.

The catch: one step at a time

- Stepwise makes the move that looks best **right now**, and never revisits it.
- It can get locked into a path that never visits the best model.
- Here its 5 predictors are exactly the best 5-predictor subset. That is luck, not a guarantee.

- Notice adjusted R^2 preferred 6 predictors and AIC prefers 5. **Different criteria, different answers**, and both curves are nearly flat from 4 to 6 predictors.

Splitting the counties

```
set.seed(2026)
test.cases    <- sample(1:nrow(counties), 50)  # about 30% of 168
test.set      <- counties[test.cases, ]
training.set <- counties[-test.cases, ]
```

`set.seed()` makes the random split reproducible: same seed, same split.

Full versus stepwise, chosen on the training set

```
full    <- lm(PhysiciansPer10000 ~ ., data=training.set)
reduced <- step(full, trace=0)

rmse <- function(actual, predicted) sqrt(mean((actual - predicted)^2))
rmse(test.set$PhysiciansPer10000, predict(full, test.set))

[1] 1.788

rmse(test.set$PhysiciansPer10000, predict(reduced, test.set))

[1] 1.824
```

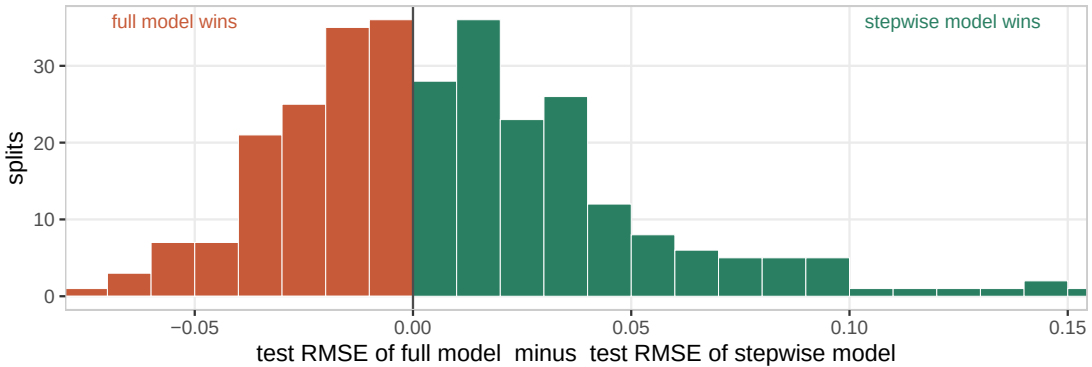
Stepwise runs on the training set too. Choosing variables with the test set would leak it into the model.

The scorecard

	Training RMSE	Test RMSE
Full model (10 predictors)	1.54	1.79
Stepwise model (4 predictors)	1.57	1.82

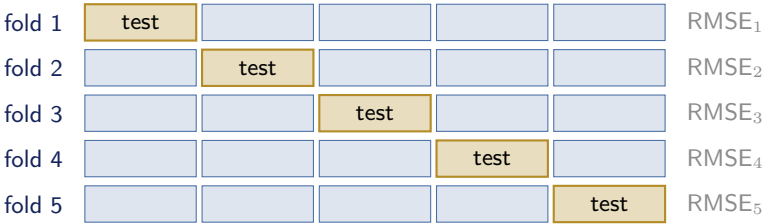
- On the training set the full model wins, as it must.
- On the test set the stepwise model does worse.

But that was one split. Here are 300.



Stepwise wins 54% of the time. One random split can tell almost any story.

Cross-validation: every case gets a turn in the test set



- Cut the data into K folds. Fit on $K-1$ of them, score on the one left out, and rotate.
- Average the K test errors. Every case is predicted exactly once, by a model that never saw it.

The verdict from 10-fold cross-validation

	10-fold CV RMSE
Full model (10 predictors)	1.68
Stepwise model (chosen inside each fold)	1.66

- Essentially a tie. With 168 counties and only 10 predictors, there is **little room to overfit**.
- Give the model more freedom (interactions, many more predictors) and that changes fast. That is where penalized regression comes in.

What to take away

- R^2 and in-sample RMSE always reward complexity. Use **adjusted** R^2 or **AIC** to charge for it.
- **Best subsets** searches everything but does not scale; **stepwise** scales but is greedy.
- The honest test is **new data**: a held-out test set, or better, cross-validation.
- Aim for the parsimonious middle, and let subject knowledge overrule a criterion when it should.